

TASK 4 :- Use various data types, list, tuple and Dictionary in python programming

(a) shopping cart price calculator (List)

AIM: TO store prices of purchased items in a list and calculate:

- Total bill amount
- Highest priced item
- Lowest priced item

Algorithm:

1. Start the program
2. Create a list to store item prices.
3. Calculate total bill using `sum()`.
4. Find the highest priced item using `max()`.
5. Find the lowest priced item using `min()`.
6. Display the results.
7. End program.

Program:

```
#shopping cart price calculator  
prices = [250, 120, 75, 300, 150]
```

```
total_bill = sum(prices)
```

```
highest_price = max(prices)
```

```
lowest_price = min(prices)
```

```
print("Total Bill Amount: ", total_bill)
```

```
print("Highest priced Item: ", highest_price)
```

```
print("Lowest priced Item: ", lowest_price)
```

SAMPLE OUTPUT

Total Bill Amount : 895

Highest priced Item : 300

lowest priced Item : 75



VEL TECH - CASE	
EX NO.	
PERFORMANCE (S)	
TEST AND ANALYSIS (S)	
RESULT AND ANALYSIS (S)	
WAVEFORM (S)	
CONCLUSION	
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10

Report: Comparison of (country) using a circuit board
in (country) & in (country) & in (country)
1/2

b) Student Exam Result (Tuple)

AIM:

TO store student names and marks in tuples, then:

- Display student with highest marks
- Display student scoring above 400

Algorithm:

1. start program
2. store data for 5 student in a list of tuples.
3. Find the student with highest marks using `max()` and a key function.
4. loop through list to display students with marks > 400 .
5. End program.

program

```
# Student Exam Result
```

```
student = [  
    ("Rahul", 456),  
    ("Shona", 389),  
    ("Amit", 412),  
    ("Priya", 478),  
    ("Kiran", 390)  
]
```

```
top_student = max(students, key = lambda x: x[1])  
print("Top Student: "; top_student[0], "with",  
      top_student[1], "marks")
```


SAMPLE OUTPUT

Top student: priya with 478 marks
students scoring above 400:

Rahul - 456

Amit - 412

Priya - 478


```
print('Student scoring above 400:')
```

```
for name, marks in students:
```

```
    if marks > 400:
```

```
        print(name, '-', marks)
```

C - Country - Capital Finder (Dictionary)

AIM:

TO store countries and capitals in a dictionary allowing:

- Adding a new entry
- Searching for a Capital
- Displaying all pairs alphabetically by country name

Algorithm .

1. start program
2. create dictionary of countries and capitals.
3. Take user input for a new country-capital pair and add to dictionary
4. ASK user for a country to search and display its capital.
5. Display all country-capital pairs sorted alphabetically.
- 6 - End program .

Sample Output

Enter country name : ** Germany **

Enter its capital : ** Berlin **

Enter country to search : ** Japan **

Capital of Japan is Tokyo

All Country - Capital Pairs

(Alphabetical):

France - Paris

Germany - Berlin

India - New Delhi

Japan - Tokyo.

program.

```
# country-capital Finder  
countries = { "India": "New Delhi", "France":  
              "Paris", "Japan": "Tokyo" }
```

```
# Add new entry  
new-country = input("Enter country name: ")  
new-capital = input("Enter its capital: ")  
countries[new-country] = new-capital
```

```
# Search for capital  
search-country = input("Enter country to search: ")  
if search-country in countries:  
    print("Capital of", search-country, "is",  
          countries[search-country])
```

```
else:  
    print("Country not found.")
```

```
# Display sorted list  
print("\n All country-capital pairs (Alphabetical):")  
for country in sorted(countries):  
    print(country, " _ ", countries[country])
```

VEL TECH - CSE	
EX NO.	14
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	1
TOTAL (20)	21
SIGN WITH DATE	13/3/18

RESULT:

Thus, various data types: list, tuples and dictionary in python programming was used and verified successfully.